MANOJ PAWAR SJ

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SUMMARY

Software Development Engineer 2 with nearly 5 years of experience building backend, distributed and full-stack systems in Go, Python, TypeScript and Node.js across space tech, e-commerce, gaming and CRM products. Currently building a real-time monitoring and control platform for ISRO's ground station network, with binary protocol parsing, a config-driven rules engine and bandwidth-optimized MQTT sync. At Digantara, made satellite collision analysis **6x faster**, cut cloud costs by **40%** and reduced manual operations work by **90%**; earlier, shipped features that raised e-commerce conversion by **5%** and CRM call pickup rates by **35%**. Strong in microservices, event-driven architecture, real-time data (MQTT, SSE, TCP/IP) and AWS/GCP.

TECHNICAL SKILLS

Languages: Go (Golang), Python, TypeScript, JavaScript, SQL
Backend: Node.js, Nest.js, Express.js, Hono, REST APIs, Microservices, WebSockets, SSE, MQTT, TCP/IP & Telnet sockets
Concepts: Distributed systems, Event-driven architecture, Pub/Sub, Real-time streaming, Rules engines, Config-driven (XML) design, Binary protocol parsing, Bit manipulation, RBAC, Caching (Redis), Concurrency (Go goroutines & channels, Python asyncio, threading, multiprocessing)
Frontend: React.js, Next.js, Tailwind CSS
Databases & ORMs: PostgreSQL, MySQL, MongoDB, Redis, Firestore, TypeORM, Drizzle, Sequelize, Mongoose
Cloud & DevOps: AWS (EC2, ECS, Lambda, SQS, S3, RDS, Route 53), Google Cloud (Compute Engine, Cloud Functions, Cloud SQL), Docker, Docker Swarm, Terraform, Gitea CI/CD, Grafana, Git, Linux (RHEL 9)
Testing: Vitest
Other: SciPy, SGP4, Azure PlayFab, Firebase, Metabase

PROFESSIONAL EXPERIENCE

Digantara Industries | digantara.co.in | Bangalore, India | 10/2023 – Present
Software Development Engineer 2 | 03/2026 – Present
Onsite contract engineer at ISTRAC, ISRO (02/2026 – 03/2027)

- Building a distributed, real-time **monitoring and control platform** for ISRO's ground station network: 4 microservices in Python and Node.js, developed on an **offline, air-gapped RHEL 9 (Linux)** network under ISRO security rules.
- **Device Integration Service:** Designed a **config-driven** Python service that connects to heterogeneous hardware over raw **TCP/IP and Telnet sockets**, parses bulk ASCII and **binary** data streams into structured data using XML-defined schemas, decoding packed fields with **bit masks, bit offsets and bitwise operations**, and sends control commands. New device types are added through configuration alone, with no code changes.
- **Rules Engine Service:** Built a Python service that aggregates and continuously polls every device service and streams live data to the backend. Designed a general-purpose, XML-based **rules engine** (AND/OR/NOT/IN expressions evaluated over live data from multiple devices) that powers alarm detection, satellite-pass readiness checks, uplink sweep checks and AOS/LOS detection. These conditions change often, so operators update them through config with **no programming knowledge**.
- **Backend API Service:** Built a Node.js service aggregating multiple rules engine instances, with real-time streaming to the UI over **SSE**, user management and **RBAC**, a command pipeline that routes user commands through the rules engine to the device layer, and publishing to external systems over **MQTT**.
- **Sync Service:** Designed a bandwidth-optimized MQTT sync protocol between site servers and a central server: on-demand subscriptions (sites publish only while a user is watching), **delta/diff calculation** to send only changed fields, periodic full snapshots, and **state reconstruction** on the central server. Remote commands use an MQTT **request/response** pattern.

Software Engineer | 10/2023 – 02/2026

- Optimized the performance of Go APIs that predict where satellites will be and check whether any two could collide, using **concurrent processing with goroutines and channels**, **cutting cloud costs by 40%** while keeping memory use low.
- Rewrote the Python collision-check algorithm using techniques from published research, cutting run time from **6 hours to 1 hour (6x faster)**: bulk batch pre-propagation of orbits with **SGP4**, relative-velocity analysis to narrow the search, and **SciPy** optimization to reliably find closest-approach minima in non-continuous functions.
- Built a job scheduler in Nest.js that lets teams change when jobs run without redeploying, along with role-based access control and user signup, **cutting manual work by 90%**.
- Turned business requirements into scalable microservices for 2 commercial products: **FDS**, software for planning satellite orbits and manoeuvres, and **SPACEMAP**, a platform that tracks objects in space; set up **Gitea CI/CD** pipelines deploying both to **AWS ECS**.
- Built the FDS web app in React.js and Tailwind CSS, with a focus on responsiveness and ease of use.

Elite Competitions | elitecompetitions.co.uk | Remote | 02/2023 – 07/2023

Software Engineer (Full Stack, Contract)

- Built a coupon system (one-time, user-specific and newsletter codes) that raised checkout conversion by **5%**.
- Set up Metabase dashboards tracking business KPIs such as customer lifetime value, cart abandonment rate and average order value.
- Built event-driven communication between services using Nest.js, Next.js, AWS Lambda and SQS.

Playshifu | playshifu.com | Bangalore, India | 09/2022 – 01/2023

Software Engineer (Full Stack)

- Built React.js dashboards for internal teams that exposed **3 major** reporting bottlenecks.
- Developed real-time backend services for multiplayer games on Azure PlayFab.

Sunstone | sunstone.in | Gurgaon, India | 01/2022 – 08/2022

Software Engineer (Backend)

- Integrated Airtel and Kaleyra cloud calling into the CRM, raising call pickup rates by **35%**.
- Built REST APIs in Express.js using MVC architecture for the CRM backend.

PROJECTS

Gobber | Go | github.com/manojpawarsj12/gobber
A Node.js package manager written in Go that installs packages significantly faster than npm.

smolie-js | TypeScript | github.com/manojpawarsj12/smolie-js
A reactive UI library under 1 KB with signals, effects and DOM rendering helpers.

Vent-Chat | Node.js, WebSockets, MongoDB
A real-time chat and social app with persistent message history.

App Tracker | Windows | github.com/manojpawarsj12/apptracker
A time-tracking app that records how long users spend on each app and website.

EDUCATION

B.E. Computer Science and Engineering | Sathyabama Institute of Science and Technology | Chennai, India
08/2018 – 07/2022 | CGPA: 8.2/10